

## LEWISTOWN 42 WSW, MT, USA

WMO: 742280

Lat: 46.8847N

Lon: 110.2895W

Elev: 1545

StdP: 84.09

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 04140

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 2             | -26.9      | -23.7 | -29.5                         | 0.3 | -25.8 | -26.6 | 0.4 | -23.3 | 11.0                  | 4.7  | 9.7 | 4.4  | 1.1                   | N/A  | 0.615 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.8                   | 30.0            | 13.8 | 28.4 | 13.4 | 26.8 | 13.1 | 15.8                | 25.3 | 15.0 | 24.9 | 14.1 | 24.3 | 3.4                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 12.2                            | 10.7 | 19.2 | 11.0 | 9.9 | 18.3 | 10.0 | 9.2 | 17.4 | 50.1          | 25.3 | 47.0 | 24.8 | 44.5 | 24.3 | 19.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.2               | 7.9  | 6.8 | -32.1                      | 32.7 | 4.0                | 1.3 | -34.9                                              | 33.6 | -37.2      | 34.4 | -39.5      | 35.1 | -42.3      | 36.0 |
|                   |      |     | -32.3                      | 18.2 | 4.2                | 0.8 | -35.3                                              | 18.8 | -37.8      | 19.2 | -40.2      | 19.7 | -43.2      | 20.3 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.9    | -3.4  | -5.8  | 0.2   | 3.2   | 7.6   | 12.4  | 16.8  | 16.0  | 11.5  | 5.0   | -0.1  | -5.3 |
|                                                                                  | DBStd       | 9.64   | 7.19  | 8.52  | 6.57  | 4.74  | 4.08  | 3.55  | 3.02  | 3.36  | 4.66  | 5.81  | 7.14  | 8.00 |
|                                                                                  | HDD10.0     | 2473   | 416   | 443   | 306   | 208   | 97    | 15    | 1     | 3     | 37    | 170   | 305   | 473  |
|                                                                                  | HDD18.3     | 4938   | 674   | 676   | 563   | 453   | 332   | 179   | 66    | 86    | 209   | 414   | 554   | 732  |
|                                                                                  | CDD10.0     | 613    | 1     | 1     | 1     | 5     | 23    | 88    | 211   | 188   | 81    | 14    | 1     | 0    |
|                                                                                  | CDD18.3     | 35     | 0     | 0     | 0     | 0     | 0     | 2     | 18    | 12    | 3     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 1371   | 0     | 0     | 0     | 1     | 9     | 113   | 565   | 496   | 175   | 12    | 0     | 0    |
|                                                                                  | CDH26.7     | 356    | 0     | 0     | 0     | 0     | 0     | 26    | 153   | 142   | 34    | 2     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.5    | 3.3   | 2.8   | 2.9   | 2.5   | 2.1   | 2.2   | 2.0   | 1.9   | 2.1   | 2.6   | 3.1   | 3.1  |
| Precipitation                                                                    | PrecAvg     | 438    | 19    | 16    | 24    | 46    | 66    | 83    | 44    | 36    | 30    | 29    | 23    | 21   |
|                                                                                  | PrecMax     | 627    | 30    | 35    | 44    | 74    | 118   | 146   | 136   | 92    | 47    | 63    | 46    | 53   |
|                                                                                  | PrecMin     | 339    | 8     | 7     | 12    | 27    | 17    | 34    | 12    | 6     | 6     | 11    | 12    | 10   |
|                                                                                  | PrecStd     | 75     | 7     | 7     | 9     | 16    | 24    | 28    | 29    | 24    | 10    | 13    | 10    | 11   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.7  | 13.8  | 17.0  | 21.9  | 24.9  | 29.8  | 31.8  | 32.4  | 29.5  | 25.1  | 18.4  | 10.7 |
|                                                                                  |             | MCWB   | 4.2   | 4.4   | 6.5   | 9.2   | 11.3  | 15.4  | 14.3  | 13.7  | 12.4  | 10.8  | 7.0   | 2.9  |
|                                                                                  | 2%          | DB     | 9.5   | 10.1  | 14.0  | 18.4  | 21.9  | 26.7  | 30.1  | 29.9  | 27.7  | 20.2  | 14.5  | 8.8  |
|                                                                                  |             | MCWB   | 2.4   | 3.1   | 5.2   | 8.0   | 10.8  | 14.2  | 14.1  | 13.4  | 11.9  | 8.7   | 5.8   | 2.4  |
|                                                                                  | 5%          | DB     | 7.7   | 7.8   | 11.9  | 15.2  | 19.9  | 23.9  | 28.5  | 28.2  | 25.4  | 17.6  | 11.4  | 6.9  |
|                                                                                  |             | MCWB   | 1.7   | 2.1   | 4.4   | 6.1   | 10.3  | 13.4  | 13.9  | 13.1  | 11.3  | 7.8   | 4.3   | 1.4  |
|                                                                                  | 10%         | DB     | 5.9   | 5.4   | 9.6   | 12.2  | 17.4  | 21.4  | 26.7  | 26.4  | 23.0  | 14.8  | 9.2   | 4.7  |
|                                                                                  |             | MCWB   | 1.0   | 0.8   | 3.1   | 4.8   | 8.9   | 12.1  | 13.6  | 12.8  | 10.7  | 6.7   | 3.2   | 0.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 4.5   | 5.1   | 7.5   | 10.2  | 13.2  | 16.9  | 17.7  | 15.7  | 15.1  | 11.8  | 7.9   | 4.1  |
|                                                                                  |             | MCDB   | 11.4  | 12.1  | 15.8  | 20.9  | 20.8  | 25.8  | 26.1  | 26.5  | 24.0  | 23.3  | 17.4  | 9.0  |
|                                                                                  | 2%          | WB     | 3.0   | 3.7   | 5.9   | 8.6   | 11.7  | 15.4  | 16.2  | 14.8  | 13.2  | 9.8   | 6.2   | 3.0  |
|                                                                                  |             | MCDB   | 8.1   | 9.8   | 13.5  | 18.4  | 19.3  | 24.2  | 25.6  | 25.8  | 24.1  | 18.2  | 13.4  | 7.8  |
|                                                                                  | 5%          | WB     | 2.0   | 2.2   | 4.6   | 6.7   | 10.7  | 14.0  | 15.3  | 14.1  | 12.2  | 8.4   | 4.8   | 1.7  |
|                                                                                  |             | MCDB   | 6.7   | 7.4   | 11.4  | 14.5  | 18.1  | 22.7  | 25.1  | 25.1  | 23.1  | 15.9  | 10.9  | 6.9  |
|                                                                                  | 10%         | WB     | 1.2   | 1.0   | 3.3   | 5.2   | 9.6   | 12.9  | 14.5  | 13.4  | 11.2  | 7.2   | 3.3   | 0.5  |
|                                                                                  |             | MCDB   | 5.9   | 5.5   | 9.0   | 11.5  | 16.3  | 20.7  | 24.7  | 24.4  | 21.8  | 14.3  | 8.8   | 5.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.2  | 11.4  | 12.1  | 12.3  | 12.7  | 14.0  | 16.8  | 17.1  | 15.8  | 12.9  | 11.0  | 10.3 |
|                                                                                  |             | MCDBR  | 12.3  | 12.3  | 15.1  | 18.0  | 18.2  | 19.3  | 20.2  | 21.2  | 21.1  | 18.2  | 13.5  | 10.8 |
|                                                                                  |             | MCWBR  | 7.2   | 7.1   | 8.3   | 9.5   | 9.3   | 9.1   | 7.8   | 8.3   | 8.9   | 8.9   | 7.5   | 6.7  |
|                                                                                  | 5% WB       | MCDBR  | 11.7  | 11.9  | 14.8  | 17.5  | 16.0  | 17.5  | 18.1  | 18.1  | 18.7  | 16.5  | 13.5  | 10.2 |
|                                                                                  |             | MCWBR  | 7.6   | 7.8   | 8.3   | 9.3   | 8.4   | 8.7   | 8.3   | 7.7   | 8.5   | 8.5   | 7.6   | 6.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.239  | 0.246 | 0.261 | 0.312 | 0.313 | 0.318 | 0.323 | 0.349 | 0.304 | 0.279 | 0.250 | 0.237 |      |
|                                                                                  | taud        | 2.486  | 2.480 | 2.543 | 2.404 | 2.478 | 2.500 | 2.509 | 2.384 | 2.502 | 2.564 | 2.548 | 2.498 |      |
|                                                                                  | Ebn at Noon | 888    | 951   | 981   | 945   | 952   | 945   | 935   | 895   | 916   | 897   | 868   | 852   |      |
|                                                                                  | Edh at Noon | 62     | 77    | 85    | 109   | 105   | 103   | 101   | 110   | 89    | 71    | 58    | 55    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.54   | 2.54  | 3.72  | 4.80  | 5.63  | 6.56  | 7.15  | 5.98  | 4.37  | 2.77  | 1.66  | 1.20  |      |
|                                                                                  | RadStd      | 0.12   | 0.17  | 0.27  | 0.28  | 0.52  | 0.45  | 0.36  | 0.33  | 0.34  | 0.23  | 0.14  | 0.09  |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (9 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | -0.55 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page